

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May - June 2023

CY1001-1 – ENGINEERING CHEMISTRY

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Function of salt bridge is**
 A) To avoid liquid junction potential
 B) To reduce polarization of the electrodes
 C) To prevent accumulation of charges around the electrodes
 D) All of the above
- If a galvanic cell uses zinc anode ($E^\circ_{\text{Zn}} = -0.76\text{V}$) and copper cathode ($E^\circ_{\text{Cu}} = 0.34\text{V}$), standard emf of the cell is**
 A) -0.42V
 B) 1.01V
 C) 1.10V
 D) -1.10V
- Which of the following is an example for ion selective electrode**
 A) Platinum electrode
 B) Gas electrode
 C) Calomel electrode
 D) Glass electrode
- Capacity of the battery is measured in**
 A) Ampere (A)
 B) Volts (V)
 C) Ampere hours (Ah)
 D) Watts (W)
- Corrosion between the dissimilar metals is called as.....**
 A) Galvanic corrosion
 B) Dry corrosion
 C) Oxidation corrosion
 D) Concentration cell corrosion
- Lower is pH, corrosion is**
 A) Greater
 B) Lower
 C) Constant
 D) None of above
- Which of the following represents correct sequence for caustic embrittlement?**
 (i) Dissolution of metal (ii) hydrolysis of sodium carbonate to form sodium hydroxide
 (iii) formation of rust (iv) trickling of alkaline water in to crevices
 A) (ii), (iv), (i), (iii)
 B) (i), (ii), (iii), (iv)
 C) (ii), (i), (iv), (iii)
 D) (iv), (ii), (i), (iii)
- The _____ potential required for continuous electrolysis is called decomposition potential.**
 A) minimum
 B) maximum
 C) constant
 D) none
- The total number of functional groups or bonding sites present in a monomer molecule is called**
 A) Functionality
 B) Degree of polymerization
 C) Molecular weight
 D) Polymer
- Which one of the following is an elastomer?**
 A) PVC
 B) Bakelite
 C) Natural Rubber
 D) Nylon
- Which of the following is a synthetic polymer?**
 A) Cellulose
 B) PVC
 C) Proteins
 D) Nucleic acids
- Guttapercha is an example for _____ polymer**
 A) Isotactic
 B) Syndiotactic
 C) Atactic
 D) Cross linked
- The purest form of water is**
 A) Rain water
 B) Lake water
 C) Sea water
 D) River water

14. Hydrolysis of MgSO_4 gives rise to
 A) $\text{MgCO}_3 + \text{H}_2\text{SO}_4$ B) $\text{Mg}(\text{OH})_2 + \text{HCl}$
 C) $\text{MgCO}_3 + \text{HCl}$ D) $\text{Mg}(\text{OH})_2 + \text{H}_2\text{SO}_4$
15. Which of the following is TRUE?
 A) Lime removes permanent hardness caused due to calcium salts B) Soda removes temporary hardness caused due to calcium salts
 C) Lime removes permanent hardness caused due to magnesium salts D) Soda removes permanent hardness caused due to magnesium salts
16. CVD stands for
 A) Carbon Vapor Dissolution B) Chemical Vapor Deposition
 C) Carbon Vapor Deposition D) Chemical Vapor Dissolution
17. Which are the main constituents of fuel from given options?
 A) Carbon and Nitrogen B) Oxygen and Hydrogen
 C) Carbon and Hydrogen D) Helium and Oxygen
18. Which gaseous fuel is found dissolved in petroleum under earth's surface in oil and gas bearing areas?
 A) Natural gas B) Coal gas
 C) Mond gas D) Producer gas
19. Thermotropic liquid crystals are obtained by varying
 A) Solvent concentration B) Temperature
 C) Pressure D) None of these
20. First liquid crystal phase discovered in a compound:
 A) Cholesteryl propanoate B) Cholesteryl benzoate
 C) Cholesteryl ethanoate D) Cholesteryl methanoate

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive Nernst equation for the following electrode reaction at standard temperature and pressure (STP): $\text{Ag}^{\oplus} + \text{e}^{\ominus} \rightleftharpoons \text{Ag}$	6	L*2	1	1
	b) Describe the construction and working of calomel electrode. Give any one application.	7	L2	1	1
	c) Calculate the EMF of the cell $\text{Ni} / \text{Ni}^{2+} (0.01\text{M}) // \text{Ag}^+ (0.05\text{M}) // \text{Ag}$ at STP. Given the standard electrode potentials of Ni and Ag are -0.23V and +0.80 respectively.	3	L2	1	1
2.	a) Explain the electrochemical theory for rusting of iron.	7	L2	1	1
	b) Discuss the following factors affecting the rate of corrosion. i) Electrode potential, ii) pH of the environment.	6	L2	1	1
	c) Explain sacrificial anode method for the prevention of corrosion.	3	L2	1	1
3.	a) Explain the following battery characteristics: i) Capacity, ii) Voltage.	6	L4	1	1
	b) Describe the construction and working of methanol oxygen fuel cell. Mention any one application	7	L2	1	1
	c) Define overvoltage. Explain any two factors affecting overvoltage values.	3	L2	1	1
Unit – II					
4.	a) Explain the free radical mechanism of addition polymerization by taking vinyl chloride as an example.	6	L2	2	1
	b) Discuss the relationship between structure and plastic deformation property of polymers.	4	L3	2	1
	c) Outline the synthesis of the following polymers: i) Polycarbonates and ii) Epoxy resin.	6	L2	2	1
5.	a) Explain the determination of total hardness of water by EDTA method.	6	L2	2	1

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| b) | Discuss the scale formation in boilers due to the presence of calcium bicarbonate in feed water. | 3 | L2 | 2 | 1 |
| c) | Describe reverse osmosis and electro-dialysis methods for desalination of water. | 7 | L2 | 2 | 1 |
| 6. a) | Define glass transition temperature. Explain any three factors affecting T _g . | 4 | L2 | 2 | 1 |
| b) | Classify the nano-materials based on dimensions. | 4 | L4 | 2 | 1 |
| c) | Describe the external treatment method of water by hot lime soda process. | 8 | L2 | 2 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain the bomb calorimetric method for the determination of calorific value of a solid fuel. | 7 | L2 | 3 | 1 |
| b) | Discuss any three reforming reactions of hydrocarbons. | 3 | L2 | 3 | 1 |
| c) | Explain the following mesophases:
i) Nematic phase, ii) Chiral nematic phase. | 6 | L2 | 3 | 1 |
| 8. a) | A coal sample with 93% C, 5% H ₂ and 2% ash is subjected to combustion in a bomb calorimeter. Calculate the gross and net calorific value of coal, given that mass of coal sample taken is 0.0095 kg, mass of water in the copper calorimeter is 2 kg, water equivalent of calorimeter is 0.7 kg, rise in temperature of water is 2.8°K and latent heat of steam is 2457.182 kJ/kg. Specific heat of water = 4.187 kJ/kg/°K. | 5 | L3 | 3 | 2 |
| b) | Describe the production of biodiesel by transesterification method. | 3 | L2 | 3 | 1 |
| c) | Explain the liquid crystalline behavior based on chemical constitution. | 8 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
